Department of Basic Science Engineering & Humanities

DISCRETE MATHEMATICS

IV Sem CSE Stream (BDMSB415C**)**

Module 1: Fundamentals of Logic

Sl.No.	Questions	Marks	CO	CL
1	Define a tautology. Prove that for any propositions p, q, r, the compound proposition $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a tautology	6	CO1	CL2
2	Construct the truth table for i) $(p \wedge (q \rightarrow r)) \rightarrow s$ ii) $[(p \rightarrow q) \wedge (p \rightarrow r)] \rightarrow (q \rightarrow r)$ iii)	6	CO1	CL2
3	Define the following: (i) Tautology and Contradiction (ii) Open statement and Quantifier Prove that for any propositions p, q, r, the following compound proposition is a tautology: $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$	6	CO1	CL2
4	Prove the following logical equivalence $(p \rightarrow q) \wedge [\sim q \wedge (r \vee \sim q)] \Leftrightarrow \sim (q \vee p)$	6	CO1	CL2
5	Prove the following logical equivalences using laws of logic $[\neg \boldsymbol{p} \wedge (\neg \boldsymbol{q} \wedge \boldsymbol{r})] \vee (\boldsymbol{q} \wedge \boldsymbol{r}) \vee (\boldsymbol{p} \wedge \boldsymbol{r}) \Leftrightarrow \boldsymbol{r}$	6	CO1	CL2
6	Test the validity of the arguments using rules of inference $p \rightarrow q \rightarrow r$ $\neg q \rightarrow \neg p$ p ——————— $\therefore r$	7	CO1	CL3
7	Test the validity of the following argument: If I study, I will not fail in the examination. If I do not watch TV in the evenings, I will study.	7	CO1	CL3

	I failed in the examination. ∴I must have watched TV in the evenings.			
8	Let p(x):$x^2 - 7x + 10 = 0$, q(x):$x^2 - 2x + 3 = 0$ and r(x):$x < 0$. Find the truth or falsity of the following statements when the universe U contains only the integers 2 and 5. If a statement is false, provide a counterexample, or explanation. (i) $\forall x, p(x) \to \neg r(x)$ (ii) $\forall x, q(x) \to r(x)$ $\exists x, q(x) \to r(x)$	7	CO1	CL3
9	Let p(x): $x \geq 0$, q(x): $x \geq 0$ and r(x): $x^2 - 3x - 4 = 0$. For the universe comprising all real numbers, find the truth values of: (i) $\exists x [p(x) \wedge q(x)]$ (ii) $\forall x [p(x) \to q(x)]$ (iii) $\exists x [p(x) \wedge r(x)]$ (iv) $\forall x [p(x) \to r(x)]$ (v) $\forall x [q(x) \to r(x)]$ (vi) $\forall x [p(x) \vee r(x)]$	7	CO1	CL3
10	Give a direct proof for each of the following: 1) For all integers' k and l, if k and l are both even then $k + l$ is even. 2) For all integers' k and l, if k and l are both even, then kl is even.	7	CO1	CL3
	Module 2: Relations and Functions			
1	a)For any non empty set A, B, C $A \times (B \cup C) = (A \times B) \cup (A \times C)$ $A \times (B - C) = (A \times B) - (A \times C)$	6	CO2	CL2
2	Let A={1,2,3,4} and let R be the relation on A defined by xRy iff x divides y .Write the elements of R write its matrix and draw diagraph. Also find in degree and out degree of each vertex	6	CO2	CL2
3	Let A={1,2,3,4,6} and 'R' be a relation on 'A' defined by aRb if and only if "a is multiple of b" represent the relation 'R' as a matrix, draw the diagraph and Relation R.	6	CO2	CL2
4	For an fixed integer $n > 1$.Prove that the relation Congruent modulo n is an equivalence relation on the set of all integers Z.	7	CO2	CL2

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5	Let S be the set of all non-zero integers and $A = SXS$ on A ,define the relation R by$(a,b)R(c,d)\ iff\ ad = bc$. Show that R is an equivalence relation.	7	CO2	CL3
6	Draw the Hasse diagram representing the positive divisors of 36	7	CO2	CL3
7	Consider the function f and g defined by $f(x) = x^3$ and $g(x) = x^2 + 1$,=$\forall x \in R$.Find $gof, fog, f^2, g^2$and also show that $fog \neq gof$.	6	CO2	CL2
8	let R: f → f be defined by $f(x) = \begin{cases} 3x - 5 & for\ x > 0 \\ -3x + 1 & for\ x \leq 0 \end{cases}$ Determine $i)\ f(0), f\left(\frac{5}{3}\right), f\left(\frac{1}{2}\right), f(1)$ $ii) f^{-1}(1), f^{-1}(-6), f^{-1}([5,5])$	6	CO2	CL3
9	Let $A = \{1,2,3,4\}\ and\ B = \{1,2,3,4,5,6\}$ i) Find how many functions are there from A to B. How many of these are one – to – one and on to functions? ii) Find how many functions are there from B to A. How many of these are one – to – one and on to functions?	7	CO2	CL3
10	Prove that if 30 dictionaries in a library contains a total 61327 pages, then at least one of the dictionaries must have at least 2045 pages.	7	CO2	CL3
Module 3: Mathematical Induction and Recursive algorithm				
1	Prove that for every positive integers $n \geq 24$ can be written as a sum of 5's and 7's.	6	CO3	CL2
2	Prove by mathematical induction that,for every positive integer,5 divides $n^5 - n$		CO3	CL3
3	Prove by mathematical induction that, for all positive integers $n \geq 1$. $1.3 + 2.4 + 3.5 + \cdots + n(n+2) = \frac{1}{6}n(n+1)(2n+7)$	6	CO3	CL2
4	By mathematical induction, Prove that $n! \geq 2^{n-1}\ \forall n \geq 1$	6	CO3	CL2
5	Prove by mathematical induction that, for all positive integers $n \geq 1$. $1 + 2 + 3 + \cdots + n = \frac{1}{2}n(n+1)$	6	CO3	CL2